

D186

Learning as a Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Before a unit test on cell parts, a teacher whose room has a large aquarium has 20 minutes for review. Last year's class liked coloring diagrams. Which use of time best supports long-term learning?

- A. Rereading the chapter twice.
- B. Highlighting key terms in the notes.
- C. A low-stakes quiz where students write what each cell part does from memory, then check answers.
- D. Coloring labeled diagrams, since students enjoy it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A student says, "I reread my notes three times and felt ready, but I blanked on the test." What best explains this?

- A. The student should have reread five times.
- B. Tests never measure real learning.
- C. The student is not a reading-style learner.
- D. Rereading felt familiar but never practiced pulling the answers from memory.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A Spanish teacher with a popular travel club has 60 minutes of vocabulary practice to schedule before a test in three weeks. Which plan will likely lead to the best long-term memory?

- A. A trip with the travel club.
- B. Six 10-minute sessions spread across the three weeks.
- C. One 60-minute session the day before the test.
- D. Two 30-minute sessions on back-to-back days in week one.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Students who crammed scored well on Friday's quiz but remembered little a month later. Classmates who spaced their practice scored slightly lower Friday but remembered much more later. What does this show?

- A.** Cramming is better, since it won on Friday.
- B.** Cramming can boost short-term scores, while spacing builds lasting memory.
- C.** Spaced practice lowers learning.
- D.** Neither method matters for memory.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Students new to solving two-step equations are struggling. The teacher, who recently switched from teaching art, has a slide with animated clip art and scrolling text next to the problem. What change most reduces unhelpful load?

- A.** Give harder problems so students struggle productively.
- B.** Add more animations to hold attention.
- C.** Remove the animation and show a step-by-step worked example before independent practice.
- D.** Never show worked examples, since students must discover methods.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A school with a new outdoor garden plans to test every student's learning style and group visual, auditory, and hands-on learners for separate lessons. The plan comes with a colorful poster set. What does research suggest?

- A.** Matching lessons to learning styles does not improve learning; choose modes that fit the content.
- B.** Only hands-on learners should use the garden.
- C.** The plan works if the style test is accurate.
- D.** The plan helps every student.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A teacher wants to teach map reading. What is the best reason to use a map-based visual activity with all students?

- A.** Only visual learners need to read maps.
- B.** Most students are visual learners.
- C.** Visual activities are always best.
- D.** Map reading is visual content, so a visual activity fits it for everyone.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

Middle schoolers in a class that meets right after lunch have lost interest in independent reading. The teacher is considering a pizza party for anyone reading 20 books. Which change best builds lasting motivation?

- A.** The pizza party for reading 20 books.
- B.** Let students choose books within their reading range and share picks in small groups.
- C.** Assign 20 books to everyone.
- D.** Move reading to the morning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A student who plays in the school band finally solves a hard proof after trying three strategies. The class is preparing for a science fair. Which teacher response best supports a growth mindset?

- A.** “Good job.”
- B.** “Great, now you'll win the science fair.”
- C.** “You kept trying new strategies until one worked. That paid off.”
- D.** “You're a natural at math.”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

After being told “You’re so smart” for weeks, a student starts avoiding harder problems. Which explanation fits the research on praise?

- A.** Praise for being smart makes mistakes feel like proof of not being smart, so the student avoids risk.
- B.** Harder problems always lower motivation.
- C.** The student has not been praised enough.
- D.** The student needs a prize for each hard problem.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher who has won a district teaching award reviews her own records and sees she sent boys of one group to the office four times as often as other students for similar behavior. She says she treats everyone the same. What is the best next step?

- A.** Rely on her award as proof of fairness.
- B.** Stop keeping referral data.
- C.** Trust her intentions, since she means to be fair.
- D.** Reflect on assumptions behind those referrals and set clear, shared criteria for responses.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

During a lively group project on local history, a student notices she is getting angry, takes three deep breaths, and returns to the task calmly. Her group won last month’s quiz bowl. Which CASEL competency is she showing?

- A.** Self-management.
- B.** Relationship skills.
- C.** Social awareness.
- D.** Self-awareness only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

Before a unit test on cell parts, a teacher ~~whose room has a large aquarium~~ has 20 minutes for review. ~~Last year's class liked coloring diagrams.~~ Which use of time best supports long-term learning?

Target word: best review activity **Decides:** Recall from memory beats review.

A	Adjacent category	Feels fluent but builds less memory.
B	Confusion set	Another review strategy, not recall.
C	Correct	Retrieval practice with feedback strengthens memory.
D	Camouflage	Enjoyment does not make it retrieval.

2. Answer D Concept 1 • transfer

A student says, "I reread my notes three times and felt ready, but I blanked on the test." What best explains this?

Target word: explain the failure **Decides:** Rereading creates familiarity, not retrieval strength.

A	Wrong stem number	More rereading repeats the weak strategy.
B	Absolute language	Overgeneralizes; the study method is the issue.
C	Confusion set	Learning styles do not explain it.
D	Correct	Familiarity is not the same as recall.

3. Answer B Concept 2

A Spanish teacher ~~with a popular travel club~~ has 60 minutes of vocabulary practice to schedule before a test in three weeks. Which plan will likely lead to the best long-term memory?

Target word: arrange practice time **Decides:** Same total time spread over days.

A	Camouflage	The club does not decide practice timing.
B	Correct	Spacing improves long-term retention.
C	Inverted operation	Massed practice fades faster.
D	Adjacent category	Still bunched, with a long gap before the test.

4. Answer B Concept 2 • transfer

Students who crammed scored well on Friday's quiz but remembered little a month later. Classmates who spaced their practice scored slightly lower Friday but remembered much more later. What does this show?

Target word: interpret the result **Decides:** Short-term vs. long-term effect.

A	Dropped element	Ignores the one-month result.
B	Correct	Spacing trades a little short-term performance for durable learning.
C	Inverted operation	It raised long-term memory.
D	Absolute language	The later results differ clearly.

5. Answer C Concept 3

Students new to solving two-step equations are struggling. The teacher, ~~who recently switched from teaching art,~~ has a slide with animated clip art and scrolling text next to the problem. What change most reduces unhelpful load?

Target word: support novices **Decides:** Novices plus extraneous clutter.

A	Inverted operation	Raises load on learners already overloaded.
B	Dropped element	Ignores that clutter uses up working memory.
C	Correct	Cuts extraneous load and guides novices.
D	Absolute language	Worked examples are most helpful for novices.

6. Answer A Concept 4

A school ~~with a new outdoor garden~~ plans to test every student's learning style and group visual, auditory, and hands-on learners for separate lessons. ~~The plan comes with a colorful poster set.~~ What does research suggest?

Target word: evaluate the plan **Decides:** Matching to style is not supported.

A	Correct	Evidence does not support style matching.
B	Camouflage	Repeats the style assumption using the setting.
C	Adjacent category	Even accurate preferences do not predict better learning when matched.
D	Absolute language	No such benefit is shown.

7. Answer D Concept 4 · transfer

A teacher wants to teach map reading. What is the best reason to use a map-based visual activity with all students?

Target word: correct basis for choosing a mode **Decides:** Match the mode to the content.

A	Inverted operation	All students need the skill.
B	Confusion set	Relies on the style label.
C	Absolute language	Some content calls for other modes.
D	Correct	The content, not a student label, decides the mode.

8. Answer B Concept 5

Middle schoolers ~~in a class that meets right after lunch~~ have lost interest in independent reading. The teacher is considering a pizza party for anyone reading 20 books. Which change best builds lasting motivation?

Target word: motivation lever **Decides:** Autonomy, competence, and relatedness.

A	Adjacent category	External rewards can crowd out interest.
B	Correct	Choice, reachable challenge, and belonging.
C	Wrong stem number	Removes choice.
D	Camouflage	Timing is not the motivation cause named.

9. Answer C Concept 6

A student ~~who plays in the school band~~ finally solves a hard proof after trying three strategies. ~~The class is preparing for a science fair.~~ Which teacher response best supports a growth mindset?

Target word: praise wording **Decides:** Praise strategy and effort.

A	Dropped element	Gives no information about what led to success.
B	Camouflage	Unrelated to the process used.
C	Correct	Credits controllable strategies and effort.
D	Inverted operation	Praises a fixed trait.

10. Answer A Concept 6 • transfer

After being told “You’re so smart” for weeks, a student starts avoiding harder problems. Which explanation fits the research on praise?

Target word: effect of trait praise — reversed from choosing praise **Decides:** Praise for a fixed trait makes challenge a threat.

A	Correct	Trait praise encourages a fixed view of ability.
B	Absolute language	Challenge can motivate when effort is valued.
C	Inverted operation	More trait praise would deepen the problem.
D	Adjacent category	A reward system does not change what praise teaches.

11. Answer D Concept 7

A teacher ~~who has won a district teaching award~~ reviews her own records and sees she sent boys of one group to the office four times as often as other students for similar behavior. She says she treats everyone the same. What is the best next step?

Target word: responding to a bias pattern **Decides:** Her own data show a pattern.

A	Camouflage	An award does not address the pattern.
B	Inverted operation	Removes the evidence rather than the bias.
C	Absolute language	Good intentions do not rule out bias.
D	Correct	Examining one’s own decisions addresses bias.

12. Answer A Concept 8

During a lively group project ~~on local history~~, a student notices she is getting angry, takes three deep breaths, and returns to the task calmly. ~~Her group won last month’s quiz bowl.~~ Which CASEL competency is she showing?

Target word: name the competency **Decides:** She regulates her own emotions.

A	Correct	Regulating one’s own emotions and behavior.
B	Adjacent category	Working with others, not regulating oneself.
C	Confusion set	Understanding others’ views and feelings.
D	Dropped element	Noticing anger is self-awareness, but calming down is self-management.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Retrieval practice	Pulling information out of memory (low-stakes quizzes, brain dumps, flashcards) strengthens learning more than rereading or reviewing notes.	Believing rereading and highlighting are the best ways to study because they feel fluent.	A choice between reviewing material and recalling it without looking.
2	Spaced practice	Spreading practice over days and weeks produces longer-lasting memory than the same amount of practice crammed together.	Thinking one long session right before a test is most efficient.	The same practice time arranged in one block vs. several sessions.
3	Cognitive load and worked examples	Working memory is limited, so novices learn best from worked examples and chunked steps; extraneous load (clutter, split attention) should be removed.	Having beginners solve complex problems with no guidance, assuming struggle alone builds learning.	New learners, a complex task, and a choice about how much guidance to give.
4	The learning-styles myth	Research does not support matching teaching to a student's preferred visual, auditory, or kinesthetic style; match the mode to the content instead.	Sorting students by learning style and teaching each group in its style.	A plan to test or label students by style.
5	Self-determination theory	Motivation grows when learners feel autonomy (meaningful choice), competence (reachable challenge), and relatedness (belonging).	Relying mainly on prizes and points to motivate, which can lower interest in tasks students already enjoy.	A motivation problem and a choice among rewards, choice, challenge, or connection.
6	Growth-mindset feedback	Praise effort, strategies, and progress rather than fixed traits like being smart, so students link success to things they can control.	Praising intelligence, which can make students avoid challenges.	Wording of praise after success or failure.
7	Examining bias for an inclusive classroom	Teachers reflect on their own assumptions, check patterns in their own decisions (who is called on, referred, or praised), and draw on students' cultures as assets.	Believing good intentions or treating everyone the same removes bias.	A pattern in a teacher's own data across student groups.
8	CASEL social-emotional competencies	The five competencies are self-awareness, self-management, social awareness, relationship skills, and responsible decision-making.	Mixing up self-management (regulating one's own emotions) with social awareness (understanding others).	A described behavior to match to a competency.

Course scope (public description): Learning as a Science examines how research from the field of learning sciences can be applied to improve teaching and learning. This course explains how teachers can create a sense of community by examining personal biases and establishing a culturally inclusive learning environment. The course also provides evidence-based strategies for improving motivation, increasing understanding and retention, and

teaching social-emotional skills that students need to be successful socially and academically. There are no prerequisites for this course.